

Thalha Ahamed

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Executive Summary

AI & Data Science undergraduate with a strong mathematical foundation and hands-on experience building applied machine learning, computer vision, and NLP systems from the ground up. Comfortable working close to model internals, from a CNN-based real-time gesture recognition pipeline (94% accuracy, ~68ms latency) to LLM and retrieval-augmented systems. Solid grounding in calculus, probability, statistics, and linear algebra through coursework and applied project work, with a strong analytical mindset and genuine interest in going deeper into systems-level and performance-focused AI engineering, including compiler frameworks and runtime optimization.

Technical Skills

- Programming: Python, C, MySQL, R
- Mathematics: Calculus, Probability & Statistics, Linear Algebra (applied through ML model development and optimization)
- AI/ML: Machine Learning, Deep Learning, Computer Vision, CNNs, NLP, Transformers, YOLO, Neural Networks
- Frameworks & Tools: TensorFlow, HuggingFace Transformers, Scikit-learn, OpenCV
- Web & Mobile: React, React Native, TypeScript
- Cloud & Databases: Google Cloud, Supabase, MySQL, Git

Soft Skills

- Problem Solving & Analytical Thinking
- Communication & Teamwork
- Ownership & Self-Direction
- Adaptability
- Time Management

Professional Experience

Southern Alloys

Role: Software Development Intern

Aug 2025 – Sep 2025

Responsibilities:

- Designed and documented system architecture and technical workflows for a Business Management Software project
- Developed and optimized interactive web applications using React with TypeScript
- Implemented scripts to automate routine processes and improve system performance
- Collaborated with the technical team to test, debug, and enhance overall application functionality

Impact:

- Delivered core modules of a business platform within 2 weeks; recognised as a dedicated, proactive, and quick learner

Ybi Foundation

Role: Data Science & AI/ML Intern

Nov 2024

Responsibilities:

- Completed a hands-on Data Science and AI/ML internship covering applied machine learning and data analysis tasks

Impact:

- Demonstrated strong analytical and problem-solving skills, completing all project tasks with a high level of professionalism

Key Projects

HOLORACT – Gesture-Based Holographic System

Tech Stack: Python, OpenCV, Computer Vision pipeline (CNN-based)

- Developed a wearable-free hand tracking system using a single standard RGB camera and CNN-based models, requiring no depth sensors or specialised hardware
- Achieved 94% gesture recognition accuracy at approximately 68ms latency and 60 FPS, tuning the pipeline for real-time performance under tight compute constraints

- Published as a peer-reviewed paper at SCRS CIMA Conference (Springer), documenting the framework's design and real-time performance results

QueryDoc – RAG-Based Document Q&A System

Tech Stack: Python, LangChain, LLM APIs, Vector Store

- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain to answer questions over custom documents
- Implemented document chunking, embedding generation, and semantic search using a vector store for context retrieval
- Integrated an LLM to generate context-grounded answers from retrieved chunks, reducing hallucination compared to standalone LLM responses

Nutrix – AI Diet Planning Startup

Tech Stack: React, TypeScript, React Native, Supabase, LLM APIs, NLP pipeline

- Architected an end-to-end AI-powered diet planning platform with web (React + TypeScript) and mobile (React Native) apps
- Designed an NLP pipeline converting free-form natural language into structured dietary profiles, extracting goals, preferences, restrictions, and health conditions
- Implemented LLM-based intent detection and entity extraction powering a recommendation engine for personalised, day-wise meal plans

Education

B.Tech – Artificial Intelligence & Data Science

B.S. Abdur Rahman Crescent Institute of Science and Technology, Chennai, India 2022 – 2026

Higher Secondary

Chettinad Vidyashram High School, Chennai, India 2010 – 2022

Awards & Accomplishments

- Publication: HOLORACT: Gesture-Controlled Holographic Framework for STEM Education – presented at SCRS CIMA Conference, published by Springer
- Internship Certification: Data Science and AI/ML – Ybi Foundation (Nov 2024)
- Smart India Hackathon (SIH) Participant
- Certification: Java (Basic) – HackerRank (Apr 2025)
- Certification: Arduino Training – Spoken Tutorial Project, IIT Bombay (Mar 2025, Score: 95%)
- Certification: English Language for Competitive Exams – NPTEL Elite, IIT Madras (Jan–Apr 2025)
- Certification: E-Business – NPTEL, IIT Kharagpur (Jul–Oct 2025)